CheXpert-Guided Multi Disease Chest X-Ray Screening Assistant

AIMI
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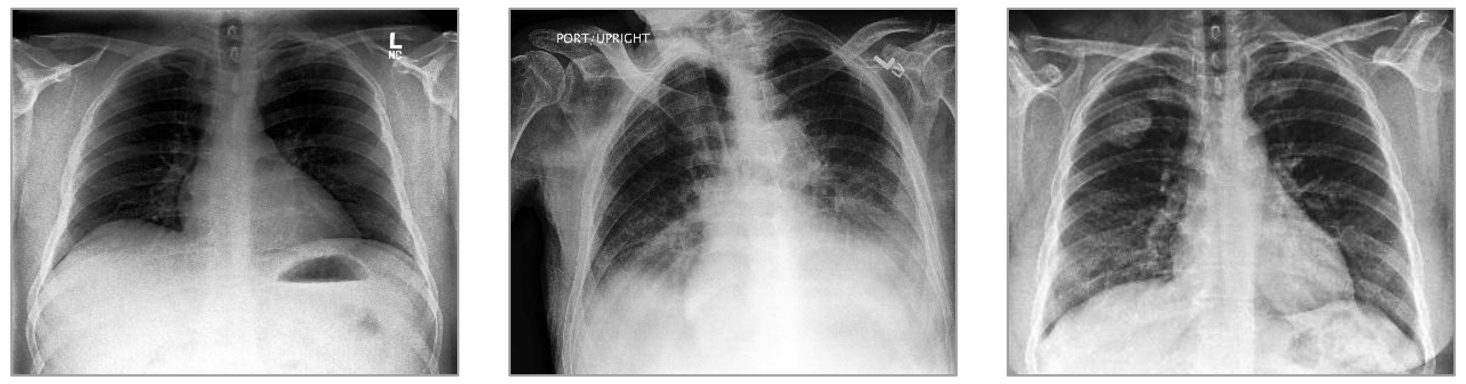
BACKGROUND

- Many chest conditions and lung diseases share **overlapping symptoms**, such as fever and wheezing
- Such similarities pose challenges for clinicians to be able to distinguish between varying thoracic conditions and correctly diagnose patients
- Existing **multi-label image classification models** often face **noisy data**, **missing labels**, and **class imbalance**, which negatively impacts patient care



Healthy Lungs **Pneumonia** **Atelectasis**

Examples of diagnosed thoracic conditions

OBJECTIVE

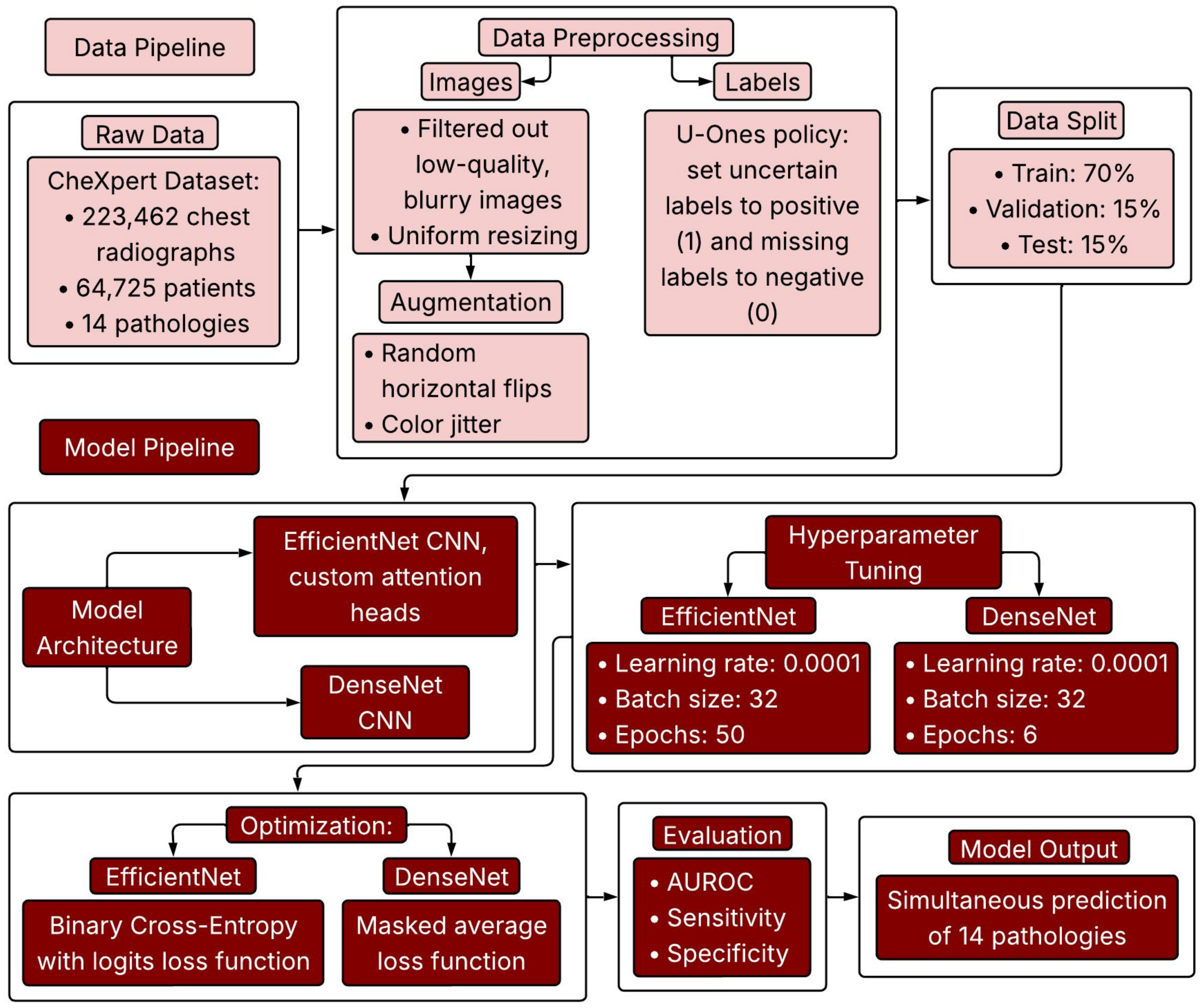
Develop a non-invasive AI-powered chest X-ray screening system designed to detect 14 distinct pathologies simultaneously

Focused on addressing two persistent limitations in current research:
(1) Weak feature localization (2) Noisy label uncertainty

DATASET

CheXpert Plus: 223,462 radiographs across 187,711 studies from 64,725 patients

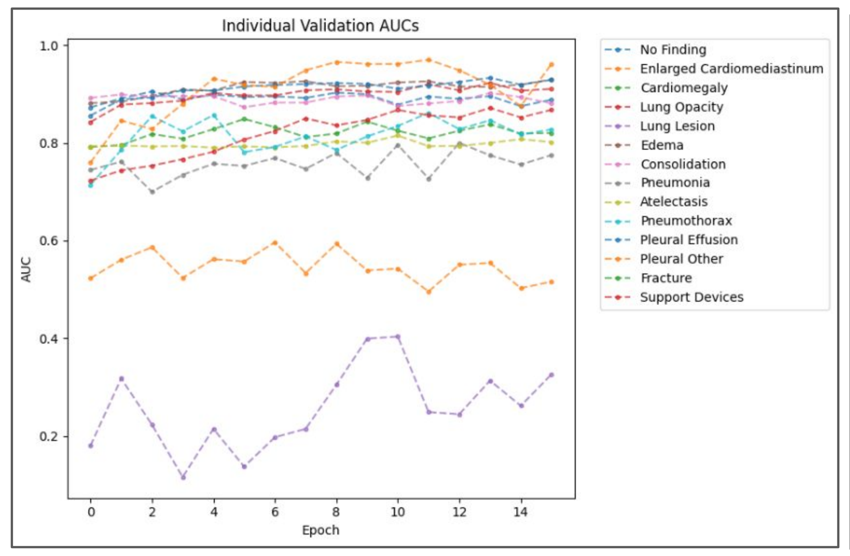
METHODS



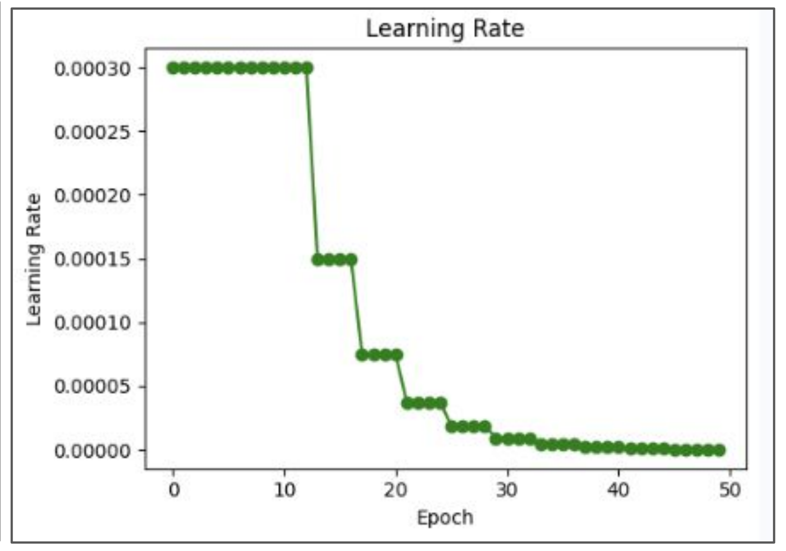
RESULTS

Method	Mean Validation AUC	Mean Sensitivity	Mean Specificity
Baseline DenseNet121	0.7952	0.7139	0.7100
Baseline EfficientNetB0	0.7948	0.7536	0.7017
DenseNet121	**0.9011 (+13.32%)**	**0.8108 (+13.57%)**	**0.8060 (+13.52%)**
EfficientNetB0	**0.8530 (+7.32%)**	**0.7801 (+3.52%)**	**0.8198 (+16.83%)**

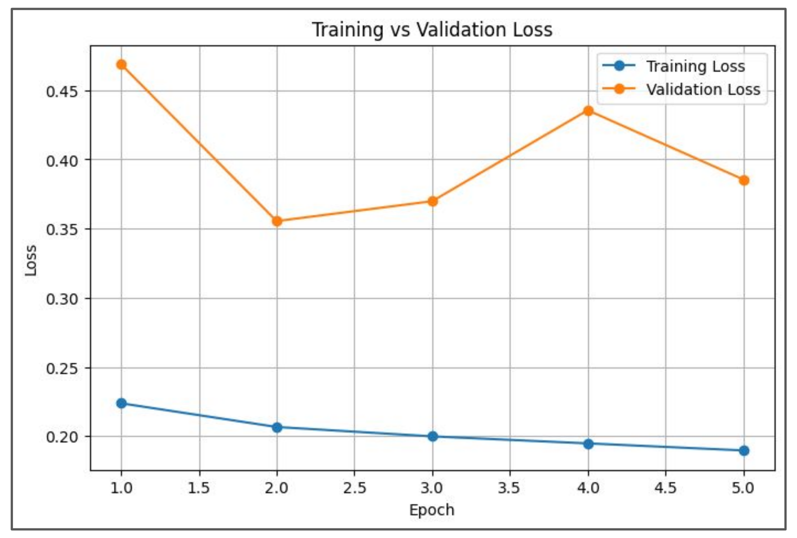
Preliminary Results: Individual AUC Scores

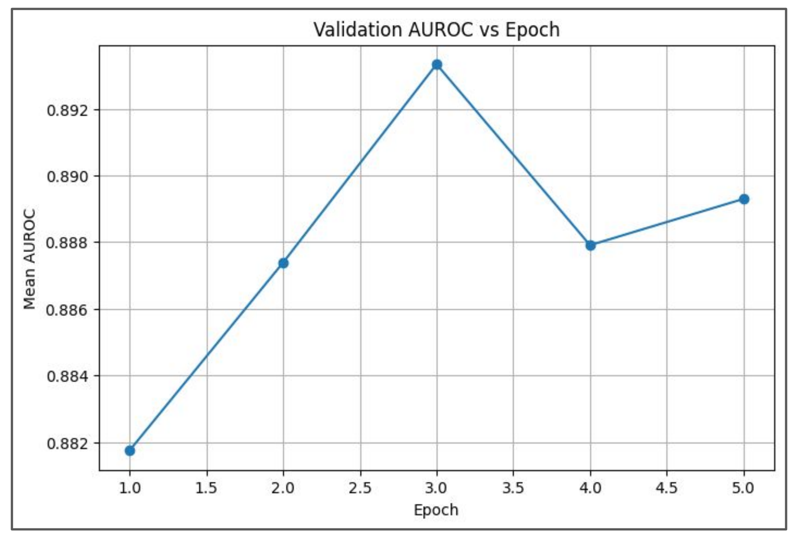


Learning Rate Reduce on Plateau

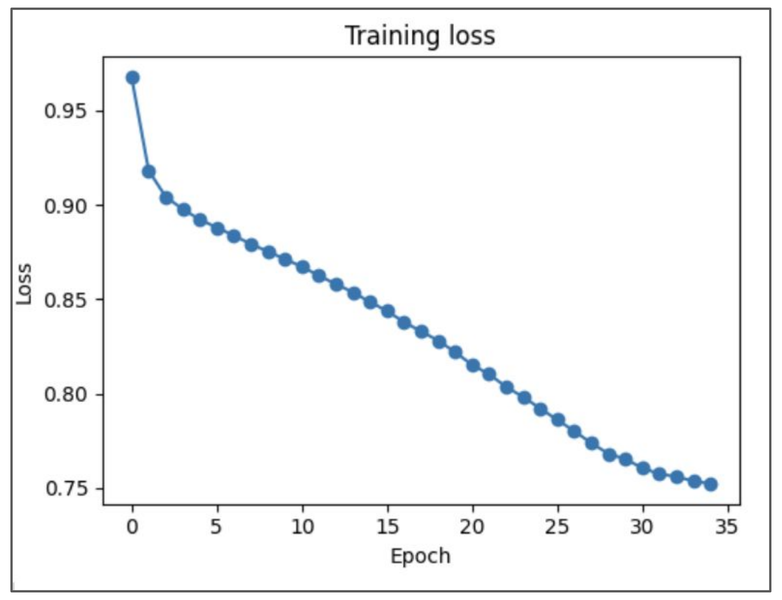


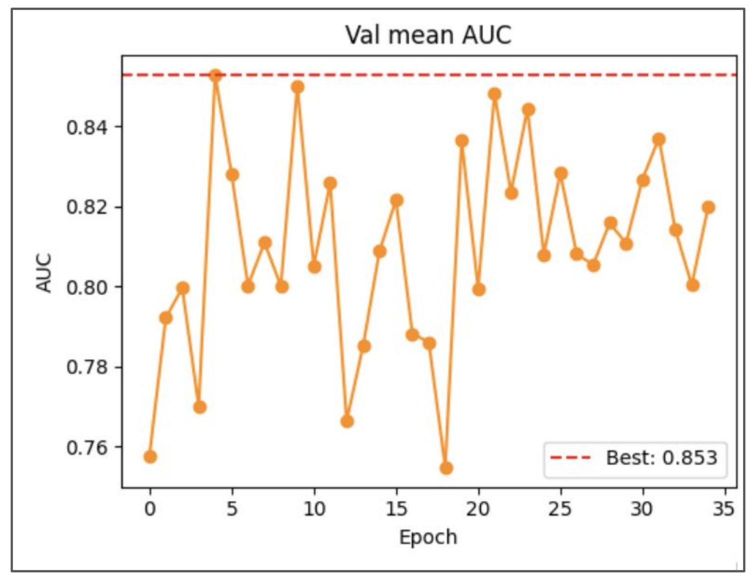
Results: DenseNet121



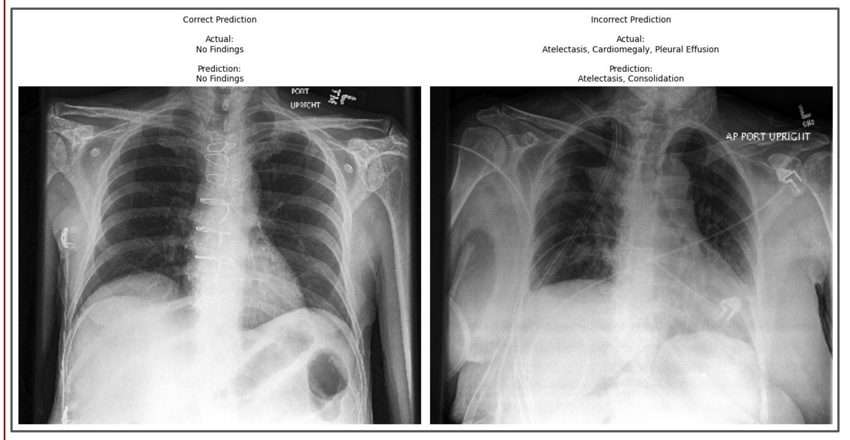


Results: EfficientNetB0

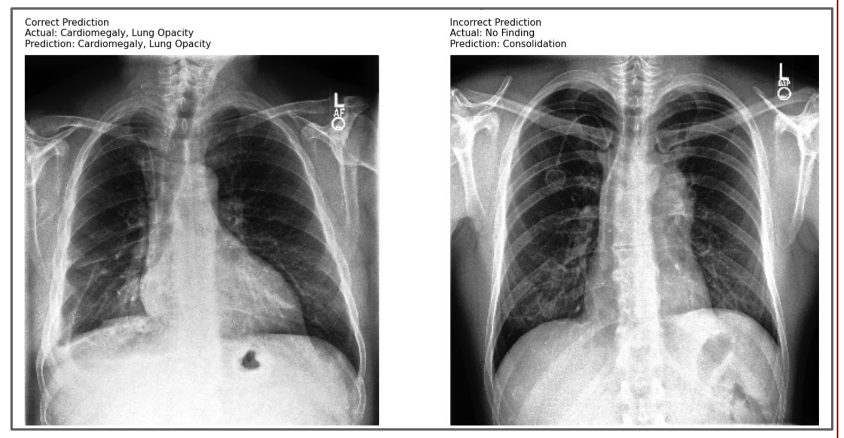




Visual Case Study: DenseNet121



Visual Case Study: EfficientNetB0



DISCUSSION

Noisy labels and highly imbalanced disease distributions in CheXpert made robust preprocessing and model selection critical for consistent and accurate diagnosis.

U-Ones Preprocessing Approach: treats uncertain labels as positive (1) and missing labels as negative (0)

Model Selection

DenseNet121	EfficientNetB0
Dense blocks with many convolutional layers connected by transition blocks	Compound scaling simultaneously scales width, depth and resolution of images

Both EfficientNetB0 and DenseNet produced strong preliminary classification performance, leading us to further model development of each.

CBAM Attention Heads (EfficientNet)
- Feature prioritization: applies channel attention (addressing which features to focus on) and spatial attention (where in the image to focus)

Multi-head self-attention: captures long-range dependencies across image

Pretrained Weights (DenseNet)
- Pretrained on ImageNet with strong recognition of edges, textures, shapes, and hierarchical visual features
- Reduces overfitting through transfer learning and robust feature representations

Implemented **early stopping with patience** and **adaptive learning rate reduction** to improve training efficiency and optimize convergence

CONCLUSION

- Performance: DenseNet121 vs. EfficientNetB0
  - EfficientNetB0: **better efficiency**
  - DenseNet121: **stronger feature detection**
  - Both struggled with visually similar pathologies and unclear patterns
- Potential for **clinical implementation** of automated multi-disease chest X-ray screenings in **high-volume healthcare settings**
- Results suggest **hybrid approaches** show promise for clinical advancement

ACKNOWLEDGMENTS

We thank our mentors, Cally Lin and Dr. Curtis Langlotz, for their guidance and support throughout this process. We acknowledge support from the Stanford AIMI Center.

REFERENCES

[1] *AIMI, S. (2026). CheXpert Plus. Redivis.com.* https://doi.org/10.57761/fzna-pm76

[2] *Das, I., Md. Alif Sheakh, Abdulla, S., Mst. Sazia Tahosin, Hassan, M. M., Zaman, S., & Shukla, A. (2025). Improving Medical X-ray Imaging Diagnosis with Attention Mechanisms and Robust Transfer Learning Techniques. IEEE Access, 1–1.* https://doi.org/10.1109/access.2025.3607639

[3] *Vendrow, E., & Schonfeld, E. (2023). Understanding transfer learning for chest radiograph clinical report generation with modified transformer architectures. Heliyon, 9(7), e17968–e17968.* https://doi.org/10.1016/j.heliyon.2023.e17968